

Library Management System

1. Problem Statement

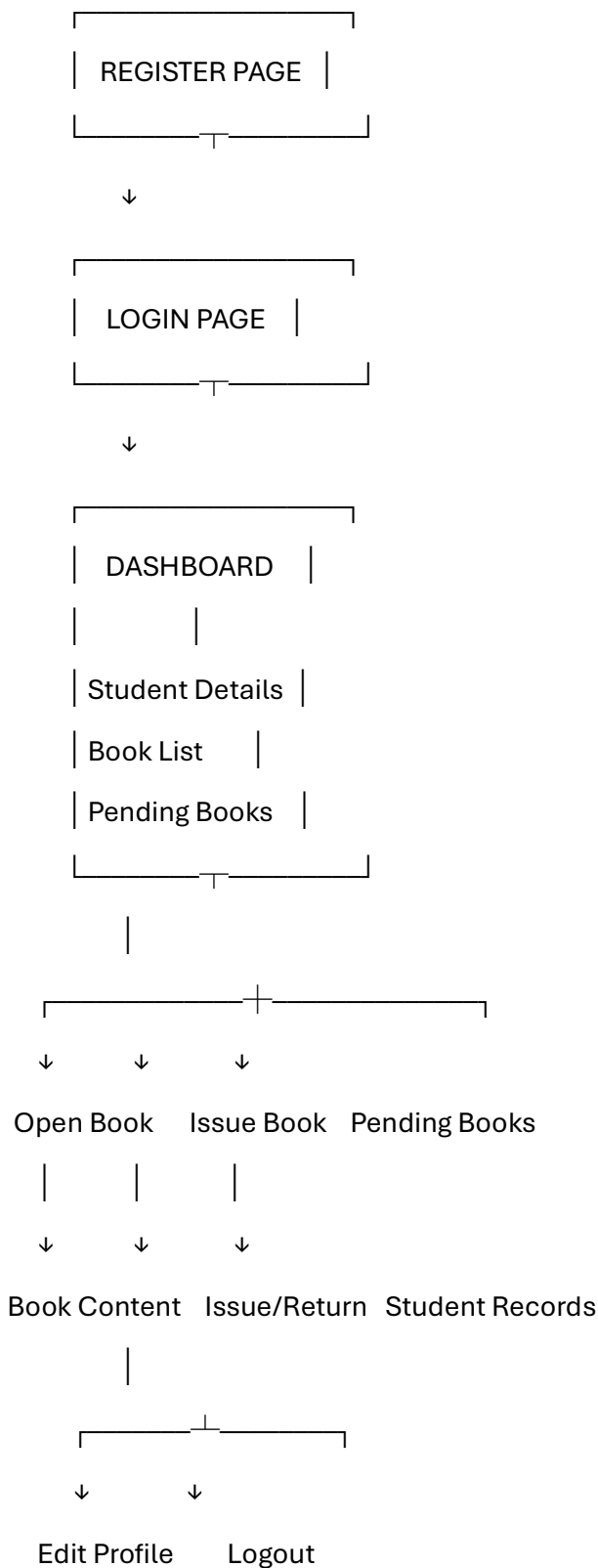
The Library Management System is designed to help students easily access and manage library books. A student can first register by entering their student ID, full name, username, and password. After registration, the student can log in to the system.

After login, the student can view the available books, book details, and their pending/issued books.

2. Actors

Actor	Role
Student	Register, login, view books, open PDF, issue/return books, edit profile, logout
Librarian/Admin	Manage books and student records
File/Database System	Stores student, book, and issued-book information
PDF Reader/Browser	Opens the selected book PDF
System Scheduler	Performs scheduled tasks such as checking pending/overdue books

3. Case Diagrame



4. Use Case Specifications

Use Case 1: Student Login

Item	Description
Use Cas	Login
Primary Actor	Student
Stakeholders	Student, Librarian/Admin
Preconditions	Student must be registered
Postconditions	Student successfully enters dashboard
Trigger	Student enters username and password and clicks Login

Main Flow

Student opens the login page.

Student enters username and password.

System validates the details.

System finds the student account.

System displays the student dashboard.

System displays books and pending books.

Alternate Flows

A1. Invalid password:

System displays "Invalid username or password" and asks the student to try again.

A2. Username not found:

System displays "User not registered" and provides an option to register.

Use Case 3: Open Book PDF

Item	Description
Use Case	Open Book
Primary Actor	Student
Stakeholders	Student, Library Administrator
Preconditions	Student must be logged in and PDF must exist
Postconditions	Selected book PDF is opened
Trigger	Student selects a book and clicks Open Book

Main Flow

Student logs into the system.

System displays the list of books.

Student selects a book.

Student clicks Open Book.

System finds the corresponding PDF file.

System opens the PDF using the default PDF reader/browser.

Alternate Flows

A1. PDF not found:

System displays "PDF file not found."

A2. PDF cannot be opened:

System displays an error message and asks the student to check the file/application.